AVERAGE AND EXTREMAL POWER-FLOW EQUATIONS

Abstract. We use probability and reinforcement learning (hopefully) to understand average and extremal behaviour of power-flow equations.

1. Introduction

Power-flow equations correspond to following collection of equations (citations here):

$$\|A_1x\|^2 = \|A_2x\|^2 = \cdots = \|A_nx\|^2 = 1$$

where A_i are $n \times n$ matrices and $x \in \mathbb{R}^n$. We are interested in understanding how many solutions this system of equations can have and the landscape of tuples of matrices $(A_1, A_2, \ldots, A_n)$ corresponding to the average and extremal behavior. Algebraic geometry results tell us that there are at most 2^n many solutions over complex numbers if the number of solutions is finitely many (cite affine Bezout exposition of Tao). However, it is unclear what are the possible number of solutions over the reals (cite $\binom{2n}{n-1}$ type result and conjectures). In this paper we aim to shed light on this using probabilistic tools and reinforcement learning.

Our scheme is as follows: We will first derive an estimate on the expected number of solutions when $A_1, A_2, \ldots, A_n$ are random Gaussian matrices with every entry centered at zero. This turns out to be $e^{\frac{n+2}{2}}$ up to constant, and we regard this as the average-case behavior. Our second step is to derive a formula for the expected number of solutions when $A_i = B_i + \delta X_i$ for each $i = 1, 2, \ldots, n$ where B_i are deterministic matrices, $\delta > 0$ is the magnitude of our perturbation, and X_i random Gaussians. This formula is a proxy to the exact count of number of solutions to the system equations

$$\|B_1x\|^2 = \|B_2x\|^2 = \cdots = \|B_nx\|^2 = 1.$$

One might wonder why we prefer a proxy instead of exact count: The reason is that current state of the art methods from computational algebra for exact counting become computationaly intractable before we can reach three digits numbers in n. Our last (mathematical) step is to use the proxy formula to create a value function for the 'state' $(B_1, B_2, \ldots, B_n)$. We will use this value function for designing policy-gradient algorithm and search for extremal systems that have the maximal 'value', that is, approximately the most number of solutions. Once these 'most-value' systems are located, our proxy can be refined for a better approximation to exact zero count, this will be discussed along the way.

2. Power Flow Equations

We model an n-node electric power network as a connected, undirected graph, $G = (V, E)$, where each vertex $v_m \in V$, $0 \leq m \leq n-1$, represents a node in the power network. There is an edge, e_{km} between vertices v_k and v_m if the corresponding nodes in the power network are connected. Each vertex has a known real valued active power injection $P_k \in \mathbb{R}$. Each edge has a known real valued susceptance $b_{km} \in \mathbb{R}$. We take $b_{km} = 0$ if the vertices k and m are not connected. We fix v_0 to be the *slack bus*. Typically we assume that slack buses have $(x_0, y_0) = (1, 0)$. In our case, we will relax this slightly and assume that the v_0 satisfies the equations

$$1 = x_0^2 + y_0^2$$
$$1 = x_0^2.$$

For $k = 1, \ldots, n-1$, the power flow equations for the network G is the system

$$P_k = \sum_{m=0}^{n-1} b_{km}(x_m y_k - x_k y_m) \tag{1}$$

$$1 = x_k^2 + y_k^2 \tag{2}$$

$$1 = x_0^2 + y_0^2 \tag{3}$$

$$1 = x_0^2. \tag{4}$$

We claim that (1)-(4) can be cast as a system of $2n$ ellipsoids in the $2n$ unknowns $x_0, \ldots, x_n, y_0, \ldots, y_n$. Consider the vector of variables $z = (x_1, \ldots, x_n, y_1, \ldots, y_n)$. Then we claim that (1)-(4) can be rewritten as $z^T A_k z = 1$ for all $1 \leq k \leq 2n$. A detailed reference is [2].

Since (1)-(4) are quadratic forms, (1) and (4) can be written as $z^T Q_k z = 1$ for some symmetric matrix Q_k and (2) and (3) can be written as $z^T S_k z = 1$ for $0 \leq k \leq n-1$.

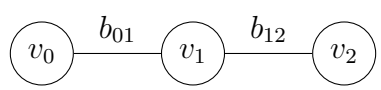
Observe that in this formulation S_k is a matrix of all zeros except in the $(k+1, k+1)$ and $(k+n, k+n)$ entries where the entry is one. In addition, $S_0 + \ldots + S_{n-1} = I_{2n}$. Therefore, instead of $z^T Q_k z = 1$ and $z^T S_k z = 1$, we define

$$\begin{aligned} A_k &= \alpha_{0,k} S_0 + \ldots + \alpha_{n-1,k} S_{n-1} \qquad \text{if } 0 \leq k \leq n-1 \\ B_k &= \beta_{0,k} S_0 + \ldots + \beta_{n-1,k} S_{n-1} + \gamma_{0,k} Q_0 + \ldots + \gamma_{n-1,k} Q_{n-1} \qquad \text{if } n \leq k \leq 2n-1, \end{aligned}$$

where $\alpha_{ij}, \gamma_{ij} \in \mathbb{R}_{>0}$ and $\beta_{ij} \in \mathbb{R}$ are chosen such that each $A_k, B_k \succ 0$ and the linear transformation they define is full rank.

Observe, in this set up, the matrices A_k, B_k inherit quite a bit of structure. Namely, each A_k is a diagonal matrix with same value in entries (i,i) and $(n+i-1, n+i-1)$. Similarly, each B_k has the same value in entries (i,i) and $(n+i-1, n+i-1)$. In addition, each B_k has an entry in the (i,j) and (j,i) entries if and only if there is an edge $e_{i+1,j+1} \in E$.

Example 2.1. *Consider the following graph on three vertices*



Here the power flow equations are

$$\begin{aligned} P_1 &= b_{01}(x_0 y_1 - x_1 y_0) + b_{12}(x_2 y_1 - x_1 y_2) \\ P_2 &= b_{12}(x_1 y_2 - x_2 y_1) \\ 1 &= x_1^2 + y_1^2 \\ 1 &= x_2^2 + y_2^2 \\ 1 &= x_0^2 + y_0^2 \\ 1 &= x_0^2 \end{aligned}$$

These equations can be written in matrix form as

$$P_1 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{b_{01}}{2} & 0 \\ 0 & 0 & 0 & -\frac{b_{01}}{2} & 0 & -\frac{b_{12}}{2} \\ 0 & 0 & 0 & 0 & \frac{b_{12}}{2} & 0 \\ 0 & -\frac{b_{01}}{2} & 0 & 0 & 0 & 0 \\ \frac{b_{01}}{2} & 0 & \frac{b_{12}}{2} & 0 & 0 & 0 \\ 0 & -\frac{b_{12}}{2} & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{b_{12}}{2} \\ 0 & 0 & 0 & 0 & -\frac{b_{12}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{b_{12}}{2} & 0 & 0 & 0 \\ 0 & \frac{b_{12}}{2} & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

$$1 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

$$1 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

$$1 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

$$1 = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}^T \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

Any linear combination of the above matrices will yield a matrix with the following sparsity pattern:

$$\begin{bmatrix} * & 0 & 0 & 0 & * & 0 \\ 0 & * & 0 & * & 0 & * \\ 0 & 0 & * & 0 & * & 0 \\ 0 & * & 0 & * & 0 & 0 \\ * & 0 & * & 0 & * & 0 \\ 0 & * & 0 & 0 & 0 & * \end{bmatrix}$$

2.1. **Maximum number of real solutions.** One question is, for a given graph G, do there exist parameter values for which all solutions are real and the total number achieves the generic complex bound. In past work, we showed that if $G = C_n$ is a cycle network on n nodes or $G = T_n$ is any tree network on n nodes, then the answer is yes [3]. An open question is whether or not this is also true for complete networks. In particular, if $G = K_4$ is the complete network on 4 nodes. In this case, the system of equations we consider is

$$\begin{aligned} P_1 &= b_{12}(x_2y_1 - x_1y_2) + b_{13}(x_3y_1 - x_1y_3) + b_{01}y_1 \\ P_2 &= b_{12}(x_1y_2 - x_2y_1) + b_{23}(x_3y_2 - x_2y_3) + b_{02}y_2 \\ P_3 &= b_{13}(x_1y_3 - x_3y_1) + b_{23}(x_2y_3 - x_3y_2) + b_{03}y_3 \\ 1 &= x_1^2 + y_1^2 \\ 1 &= x_2^2 + y_2^2 \\ 1 &= x_3^2 + y_3^2. \end{aligned}$$

In particular, we are interested in the case when $P_1 = P_2 = P_3 = 0$. The question is then, do there exist choices of $b_{01}, b_{02}, b_{03}, b_{12}, b_{13}, b_{23} \in \mathbb{R}$ such that the above system of equations has 20 real, isolated solutions? We have instances with 18 real solutions but have not been able to find an instance with 20. In case it is helpful for training, I included a file `K4_susceptances_18.rtf` with gives tuples $(b_{01}, b_{02}, b_{03}, b_{12}, b_{13}, b_{23})$ where there are 18 real solutions.

3. Probabilistic Setup and Rice Formula

We prefer a strange beginning and a happy ending. So, we start with stating our technical tool which is Theorem 6.7 in Level Sets and Extrema of Random Fields and Processes by Azais and Wschebor `https://www.math.univ-toulouse.fr/~azais/styles/other/student/level.pdf`, published version is by Wiley. The result requires some assumptions:

(1) We have random field $Z(t) \in \mathbb{R}^d$ that is created as $Z(t) = H[Y(t)]$ where
$$\{Y(t) : t \in W\}$$
is a Gaussian field indexed by a set $W \in \mathbb{R}^d$ and having values $Y(t) \in \mathbb{R}^n$ for some $n \geq d$, and $H : \mathbb{R}^n \to \mathbb{R}^d$ is a continuously differentiable function.
(2) $Z(t)$ has a density $p_{Z(t)}(x)$ that is continuous function of the pair $(t, x) \in W \times \mathbb{R}^d$
(3) $\mathbb{P}\{\exists t \in W : Z(t) = u \ , \ \det(Z'(t)) = 0\} = 0$

Then, we have

$$\mathbb{E}\left(N_u(Z, B)\right) = \int_B \mathbb{E}\left(|det(Z'(t))| \mid Z(t) = u\right) \ p_{Z(t)}(u) \ dt \tag{5}$$

where $N_u(Z, B)$ denotes the number of points in B that satisfies $Z(t) = u$, and we assume B is a Borel set.

3.1. **Random Matrices with Entries Centered at Zero.** For $A_1, A_2, \ldots, A_n$ independent $n \times n$ random matrices that have independent Gaussian entries that are centered at 0 and standard deviation σ, we are interested in counting the expected number of real solutions to the following collection of equations:

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \cdots = \|A_n x\|^2 = 1. \tag{6}$$

So, our random fields is $Z(x) = H[Y(x)]$ is indexed in $W = \mathbb{R}^n$, and $Y(x)$ is given by

$$Y : \mathbb{R}^n \to \mathbb{R}^{n^2} \ , \ Y(x) = (A_1 x, A_2 x, \ldots, A_n x)$$

Then our map H on $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^{n^2}$ where $y_i \in \mathbb{R}^n$ for all $i = 1, 2, 3, \ldots, n$ is defined as follows:

$$H : \mathbb{R}^{n^2} \to \mathbb{R}^n \ , \ H(y_1, y_2, \ldots, y_n) = (\|y_1\|^2, \ldots, \|y_n\|^2)$$

where $\|.\|$ denotes the standard Euclidean norm. We define $Z(x) := H \circ Y(x)$. Now, using the Rice formula provided by Azais and Wschebor in Equation (5) for $u = (1, 1, 1, \ldots, 1)$ we have

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) = \int_{\mathbb{R}^n} \mathbb{E}\left(|det(Z'(x))| \mid Z(x) = u\right) \ p_{Z(x)}(u) \ dx \tag{7}$$

Now we need to unravel the density $p_{Z(x)}(u)$:

$$p_{Z(x)}(1, 1, \ldots, 1) = p_{H(Y(x))}(1, 1, \ldots, 1) = P_{H(Y(\|x\| e_1))}(1, 1, \ldots, 1)$$

Where the second equality follows from rotational invariance of the Gaussian field Y. Now note that $A_1, A_2, \ldots, A_n$ are independent, so we have

$$P_{H(Y(\|x\| e_1))}(1, 1, \ldots, 1) = P_{H(Y\|x\| e_1)}(1, \ldots, 1) = \left(P_{\|x\|^2 \|A_1 e_1\|^2}(1)\right)^n$$

The random variable $\|x\|^2 \|A_1 e_1\|^2 = \|x\|^2 \left(\psi_1^2 + \ldots + \psi_n^2\right)$ where $\psi_i \sim \mathcal{N}(0, \sigma^2)$, and $\frac{\|x\|^2 \|A_1 e_1\|^2}{\|x\|^2 \sigma^2}$ follows a χ_n^2 distribution. So we have

$$P_{\|x\|^2 \|A_1 e_1\|^2}(1) = P_{\chi_n^2}\left(\frac{1}{\sigma^2 \|x\|^2}\right) \frac{1}{\|x\|^2 \sigma^2}$$

and

$$p_{Z(x)}(1, 1, \ldots, 1) = \left(\frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \left(\frac{1}{\sigma^2 \|x\|^2}\right)^{\frac{n}{2} - 1} e^{-\frac{1}{2\sigma^2 \|x\|^2}} \frac{1}{\|x\|^2 \sigma^2} \right)^n . \tag{8}$$

3.2. **Random Matrices with entries not centered at zero.** We recall the following basic fact to motivate our discussion:

Lemma 3.1 (Barvinok Partition Functions book Thm 4.5.9)**.** *Let $M_1, M_2, \ldots, M_n$ be positive definite matrices, then there exists an invertible matrix Z and positive scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that $T_i = \alpha_i Z^T M_i Z$ satisfies*

$$\text{Trace}(T_i) = 1 \textit{ for each } i = 1, 2, \ldots, n, \textit{ and } T_1 + T_2 + \ldots + T_n = \mathbb{I}$$

We are interested in counting the number of real solutions to the following system of equations:

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \ldots = \|A_n x\|^2 = 1 \text{ for } i = 1, 2, \ldots, n. \tag{9}$$

It is clear that the number of solutions to Equation (9) is equal to the number of solutions to Equation (10)

$$\|A_1 Zx\|^2 = \|A_2 Zx\|^2 = \ldots = \|A_n Zx\|^2 = 1 \text{ for } i = 1, 2, \ldots, n. \tag{10}$$

for any invertible matrix Z. Also note that scaling all equations at once does not change the number of solutions. Using Lemma 3.1, and noting $\text{Trace}(Z^T A_i^T A_i Z) = \|A_i Z\|_F^2$, we can find a Z such that

$$\frac{1}{\|A_1 Z\|_F^2} Z^T A_1^T A_1 Z + \frac{1}{\|A_2 Z\|_F^2} Z^T A_2^T A_2 Z + \ldots + \frac{1}{\|A_n Z\|_F^2} Z^T A_n^T A_n Z = \mathbb{I}$$

Note that for any matrix X solving

$$\|Xx\|^2 = 1 \text{ is equivalent to solving } \|\frac{1}{\|X\|} Xx\|^2 = \frac{1}{\|X\|^2}$$

So, we define $C_i = \frac{1}{\|A_i Z\|} A_i Z$ Equation (9).

$$\|C_i\| = 1 \text{ for each } i, \text{ and } C_1^T C_1 + C_2^T C_2 + \ldots + C_n^T C_n = \mathbb{I}$$
$$c_i = \frac{1}{\|A_i Z\|^2} \text{ for } i = 1, 2, \ldots, n$$
$$\text{Count } x \in \mathbb{R}^n \text{ such that } \|C_i x\|^2 = c_i$$

Now to make our computations easier, we do the following:

(1) Compute $g := \left(\frac{1}{\|A_1 Z\|^2} + \frac{1}{\|A_2 Z\|^2} + \cdots + \frac{1}{\|A_n Z\|^2} \right)^{\frac{1}{2}} = (c_1 + c_2 + \cdots + c_n)^{\frac{1}{2}}$.

(2) Do a change of variables by setting $x = \frac{\sqrt{n}}{g} x$. After this change of variables, our equations become $\|C_i x\|^2 = \frac{n c_i}{g^2}$.

In summary, we defined

$$C_i = \frac{1}{\|A_i Z\|} A_i Z \ , \ g = \left(\sum_{i=1}^{n} \frac{1}{\|A_i Z\|^2} \right)^{\frac{1}{2}}$$

Now we update our definition of c_i as follows:

$$c_i = \frac{n}{g^2 \|A_i Z\|^2}$$

Then our goal is the following:

$$\begin{aligned} \|C_i\| = 1 \text{ for each } i, \text{ and } C_1^T C_1 + C_2^T C_2 + \ldots + C_n^T C_n = \mathbb{I} \\ (c_1, c_2, \ldots, c_n) \in (\mathbb{R}_+)^n \text{ and } c_1 + c_2 + \cdots + c_n = n \\ \text{Count } x \in \mathbb{R}^n \text{ such that } \|C_i x\|^2 = c_i. \end{aligned} \tag{11}$$

Our model of randomness is set-up as follows: We use $c = (c_1, c_2, \ldots, c_n)$ to denote our n-tuple with $c_i > 0$ and $c_1 + c_2 + \ldots + c_n = n$, and $\mathcal{B} = (B_1, B_2, \ldots, B_n)$ such that $B_1^T B_1 + B_2^T B_2 + \ldots + B_n^T B_n = \mathbb{I}$. Then we have random Gaussian matrices A_i that are defined as

$$\begin{aligned} A_i = B_i + \delta X_i \text{ where } X_i \text{ have independent } \mathcal{N}(0,1) \text{ entries} \\ \|A_1 x\|^2 = c_1 \ , \ \|A_2 x\|^2 = c_2 \ , \cdots , \ \|A_n x\|^2 = c_n \end{aligned} \tag{12}$$

We will use the expected number of solutions to Equation (12) as a proxy to exactly counting the number of real solutions of Equation (11).

Now suppose we have solution $x \in \mathbb{R}^n$ to Equation (12), then

$$n = \sum_{i=1}^{n} x^T A_i^T A_i x = \|x\|^2 + x^T \left(\delta \sum_{i=1}^{n} (X_i^T B_i + B_i^T X_i) + \delta^2 \sum_{i=1}^{n} X_i^T X_i \right) x \tag{13}$$

Note that

$$X := \delta \sum_{i=1}^{n} (X_i^T B_i + B_i^T X_i) + \delta^2 \sum_{i=1}^{n} X_i^T X_i$$

is a real symmetric matrix, and we have

$$n = \|x\|^2 + x^T X x.$$

Moreover, $X \sim \delta^2 n \, \mathbb{I}$ with high probability. Therefore, with high probability, we have

$$n \sim (1 + \delta^2 n)\|x\|^2. \tag{14}$$

Following our notation in Section 3.1, we define the maps Y, H and Z the same way, and the same formula Equation (7) holds true. To ease the computation of this formula, specifically the computation of the conditional expectation, we will use a trick invented in [1] alongside with other tricks we invented: These will be explain in Section 5.

4. Expected Number of Common Intersections for Random Matrices with Entries Centered at Zero

First note that $Z(x) = (\|A_1 x\|^2, \|A_2 x\|^2, \ldots, \|A_n x\|^2)$ and we need to compute $Z'(x)$. For convenience let us define the following map that sends $x \in \mathbb{R}^n$ to an $n \times n$ matrix $A(x)$ as follows:

$$A : \mathbb{R}^n \to \mathbb{R}^{n \times n} \ , \ A(x)^T = [A_1^T A_1 x, A_2^T A_2 x, \ldots, A_n^T A_n x]$$

Then, we have

$$Z'(x) = 2A(x) \ , \ Z(x) = A(x)x$$

Now let $y = Qx$ for an orthogonal matrix Q, define $B_i = A_i Q$, and also extend this definition to the definition of the map $B(x)$ as

$$B(x)^T = [B_1^T B_1 x, \ldots, B_n^T B_n x].$$

Then, we have $A(y)y = B(x)x$, but $A(y) \neq B(x)$. However, we note that every row of $B(x)$ is obtained acting on rows of $A(y)$ with the orthogonal matrix Q. So, we naturally have $\det(B(x)) = \det(A(y))$. Now note that rows and columns of A_i are centered Gaussian random vectors, therefore due rotational invariance of Gaussians the distribution of A_i does not change under the action of Q. This means the following:

$$\mathbb{E}\left(|det(Z'(x))| \mid Z(x) = (1, 1, \ldots, 1)\right) = \mathbb{E}\left(|det(Z'(y))| \mid Z(y) = (1, 1, \ldots, 1)\right)$$

Also note that the density formula Equation (8) is invariant under the action of Q as well. This means using an arbitrary choice $x = e_0$ we can simplify Equation (7) as follows:

$$|S^{n-1}| \int_0^\infty \mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1, \ldots, 1)\right) \, p_{Z(te_1)}(1, \ldots, 1) t^{n-1} \, dt \tag{15}$$

where $|S^{n-1}|$ denotes the volume of the sphere S^{n-1}. Note that

$$Z(te_1) = (1, \ldots, 1) \Leftrightarrow Z(e_1) = (\frac{1}{t^2}, \ldots, \frac{1}{t^2}) = \left((A_i A_i^T)_{11}\right)_i, \text{ and } Z'(te_1) = tZ'(e_1).$$

Therefore,

$$\mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1,\dots,1)\right) = t^n\, \mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (\frac{1}{t^2},\dots,\frac{1}{t^2})\right)$$

Let the columns of the matrices A_i be a_{ij} for $j = 1, 2, \dots, n$. The assumption $Z(e_1) = (\frac{1}{t},\dots,\frac{1}{t^2})$ is equivalent to $\|a_{i1}\| = \frac{1}{t}$ for all i. Moreover, we have

$$Z'(e_1) = 2\begin{bmatrix} \langle a_{11}, a_{11}\rangle & \langle a_{11}, a_{12}\rangle & \dots & \langle a_{11}, a_{1n}\rangle \\ \vdots & & & \vdots \\ \langle a_{n1}, a_{n1}\rangle & \langle a_{n1}, a_{n2}\rangle & \dots & \langle a_{n1}, a_{nn}\rangle \end{bmatrix}$$

That is, the assumption $Z(e_1) = (\frac{1}{t},\dots,\frac{1}{t^2})$ vs $Z(e_1) = (1,\dots,1)$ scales the determinant with $\frac{1}{t^2}\cdot\frac{1}{t^{n-1}} = \frac{1}{t^{n+1}}$. This yields,

$$\mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1,\dots,1)\right) = t^{-1}\mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1,\dots,1)\right).$$

So, we can further simplify Equation (15) as

$$|S^{n-1}|\, \mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1,\dots,1)\right) \int_0^\infty t^{n-2} p_{Z(te_1)}(1,\dots,1)\, dt \tag{16}$$

The following proposition deserves a proof. To be done soon

Proposition 4.1.

$$|S^{n-1}|\, \mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1,\dots,1)\right) \sim c\, \pi^{\frac{n-1}{2}} 2^{n+1} e^{\frac{n}{2}} \sigma^{n-1}\sqrt{n}$$

Proof. (Idea for a proof) Note that assuming $Z(e_1) = (1, 1, \dots, 1)$ corresponds to that the variance of every entry in the first vector of every matrix A_i is $\frac{1}{n}$. This also means the first column of $Z'(e_1)$ is $(2, 2, \dots, 2)$. Let Π denote the projection on the orthogonal complement of the line spanned by the vector $(1, 1, \dots, 1)$. Let $z_2, z_3, \dots, z_n$ denote the second, third, and so on up to n'th column of $Z'(e_1)$. We observe that

$$\mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1,\dots,1)\right) = \sqrt{n}\mathbb{E}|\det([\Pi(z_2), \Pi(z_n), \dots, \Pi(z_n)])|.$$

Now note that the variance of every entry in the vectors $\Pi(z_2)$ is σ^2. and the random matrix $[\Pi(z_2), \Pi(z_n), \dots, \Pi(z_n)]$ obeys the assumptions of the main result in `https://arxiv.org/pdf/1112.0752.pdf`. So, we should have

$$\mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1,\dots,1)\right) = c_1 2^n \sigma^{n-1} e^{\frac{n\ln n}{2}}$$

Note that

$$|S^{n-1}| = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n}{2})} \sim 2\pi^{\frac{n-1}{2}} e^{\frac{-(n-1)\ln n + n}{2}}$$

where $c_2 < 1$. ■

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) \sim c\, \pi^{\frac{n-1}{2}} 2^{n+1} e^{\frac{n}{2}} \sigma^{n-1}\sqrt{n} \int_0^\infty t^{n-2} p_{Z(te_1)}(1,\dots,1)\, dt \tag{17}$$

$$p_{Z(te_0)}(1,\dots,1) = \left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} (\sigma t)^{-n^2} e^{-\frac{n}{2\sigma^2 t^2}}$$

$$\sigma^{n-1}\int_0^\infty t^{n-2} p_{Z(te_0)}(1,\dots,1)\, dt = \left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} \int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\, dt$$

For $u = \frac{1}{\sigma t},\ du = -\frac{1}{\sigma t^2}\ dt$ we have

$$\int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\ dt = \int_0^\infty u^{n^2-n} e^{-\frac{nu^2}{2}}\ du$$

Notice we are having a Gaussian moment here, that is, for $X \sim \mathcal{N}(0, \frac{1}{n})$ we have

$$\mathbb{E}[|X|^{n^2-n}] = \frac{2\sqrt{n}}{\sqrt{2\pi}} \int_0^\infty u^{n^2-n} e^{-\frac{nu^2}{2}}\ du = (\frac{1}{\sqrt{n}})^{n^2-n} \frac{2^{\frac{n^2-n}{2}} \Gamma(\frac{n^2-n+1}{2})}{\sqrt{\pi}}$$

Hence, we have

(18)
$$\left(2^{\frac{n}{2}} \Gamma(\frac{n}{2})\right)^{-n} \int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\ dt = \left(2^{\frac{n}{2}} \Gamma(\frac{n}{2})\right)^{-n} n^{-\frac{n^2-n+1}{2}} 2^{\frac{n^2-n-1}{2}} \Gamma(\frac{n^2-n+1}{2})$$

This gives the following,

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) = c\ \pi^{\frac{n-1}{2}} n^{-\frac{n^2-n}{2}} 2^{\frac{n+1}{2}} e^{\frac{n}{2}} \Gamma(\frac{n}{2})^{-n} \Gamma(\frac{n^2-n+1}{2}) \tag{19}$$

Recall that (Stirling):

$$\Gamma(z) \sim \sqrt{2\pi} e^{-z} z^{z-\frac{1}{2}}$$

This implies,

$$\Gamma(\frac{n}{2})^{-n} \Gamma(\frac{n^2-n+1}{2}) \sim (2\pi)^{\frac{-n+1}{2}} e^{\frac{n-1}{2}} (\frac{n}{2})^{-\frac{n^2-n}{2}} (\frac{n^2-n+1}{2})^{\frac{n^2-n}{2}}$$

$$\Gamma(\frac{n}{2})^{-n} \Gamma(\frac{n^2-n+1}{2}) \sim (2\pi)^{\frac{-n+1}{2}} e^{\frac{n-1}{2}} \left(n - 1 + \frac{1}{n}\right)^{\frac{n^2-n}{2}}$$

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) \sim c\ 2e^{\frac{2n-1}{2}} (1 - \frac{1}{n} + \frac{1}{n^2})^{\frac{n^2-n}{2}} \tag{20}$$

Note that

$$(1 - \frac{1}{n} + \frac{1}{n^2})^{\frac{n^2-n}{2}} \sim \left((1 - \frac{1}{n})^n\right)^{\frac{n-1}{2}} \sim e^{-\frac{n-1}{2}}$$

So, we have

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) = 2\ c_0\ e^{\frac{n}{2}}$$

where c_0 is a universal constant.

Theorem 4.2. *Let $A_1, A_2, \ldots, A_n$ be random Gaussian matrices with independent entries of standard deviation $\sigma > 0$. Then, the expected number of solutions to the following system of equations*

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \ldots = \|A_n x\|^2 = 1$$

is

$$c_1 e^{\frac{n+2}{2}}.$$

Julia Lindberg performed an experiment as follows. She creates samples of random $n \times n$ Gaussian matrices $A_1, \ldots, A_n$ then does the following: Computes $\|A_i^T A_i\|$ for all $i = 1, 2, \ldots, n$ and then creates the matrices

$$\frac{A_i^T A_i}{\|A_i^T A_i\|} \text{ for } i = 1, 2, \ldots, n.$$

Then she solves the corresponding equations and counts the number of real zeros. Then she computes the average number of real zeros for $2 < n \leq 10$. This data and model fitting Jupyter notebook is included in our repo. If we look at the fit in Jupyter notebook based on Julia's data, it says the fit is

$$2.0009e^{0.352n} = 2.0009(1.4219)^n \sim (2.02)^{\frac{n+2}{2}}$$

Note that this experiment is different than our model of randomness in this note.

5. Computation of Integral formula for random ellipsoids with prescribed centers

We use the notation from Section 3.2: $\mathcal{B} = (B_1, B_2, \ldots, B_n)$ denotes collection matrices that satisfies

$$\|B_i\|_F = 1 \text{ for each } i = 1, 2, \ldots, n \ , \ B_1^T B_1 + B_2^T B_2 + \ldots + B_n^T B_n = \mathbb{I}$$

and $c = (c_1, c_2, \ldots, c_n)$ a tuple of positive numbers with $c_1 + c_2 + \ldots + c_n = n$. Then we define $A_i = B_i + \delta X_i$ where X_i are Gaussian random matrices with indepedent $\mathcal{N}(0, 1)$ entries, $\delta > 0$ is our perturbation parameter.

5.0.1. *How to normalize the equations?* We initiated a normalization of power-flow equations without changing the number of their solutions in Section 3.2. Here we explain how to do the normalization algorithmically. Vinny, please explain the algorithms and implementation you have for this. Also put your nice graphs. If I remember right, you were going to add a Newton corrector on top of your implementation to make it even better.

5.0.2. *Settin up Kac-Rice Formula.* In this setting, we denote the average number of solutions to the system of equations

$$\|A_1 x\|^2 = c_1 \ , \|A_2 x\|^2 = c_2 \ , \ \cdots \ , \ \|A_n x\|^2 = c_n$$

with $\mathbb{E}N(\mathcal{B}, \delta, c)$. Following the discussion and notation in Section 3.2, we have

$$\mathbb{E}N(\mathcal{B}, \delta, c) = \int_{\mathbb{R}^n} \int_{K_c(x)} |\det Z^{'}(x)| \ d\mu(X_1, X_2, \ldots, X_n) \ dx.$$

Now recall that the weight $\mu(K_c(x))$ is very small unless x satisfies Equation (14), that is we must have

$$\|x\| \sim \frac{\sqrt{n}}{\sqrt{1 + \delta^2 n}}.$$

We should also note that for $A_i = B_i + \delta X_i$ to serve as decent proxy to B_i we must have

$$\delta n = \mathbb{E}\|\delta X_i\|_F < \|B_i\| = 1 \Rightarrow \delta < \frac{1}{n}.$$

So, we restrict $0 < \delta \leq \frac{1}{n}$ and in this parameter regime we must have $\|x\| \sim \sqrt{n}$ in order for $\mu(K(x))$ to be non-negligible. We will add an $\varepsilon > \frac{4}{\sqrt{n}}$ tolerance, and we will aim to sample vectors uniformly from the annulus

$$(1-\varepsilon)\sqrt{n} < \|x\| < (1+\varepsilon)\sqrt{n}$$

Here we recall two facts about the Gaussian measure: (1) Gaussian measure is rotationally invariant, (2) For standard Gaussian vector $x \in \mathbb{R}^n$, we have

$$\mathbb{P}\{(1-\varepsilon)\sqrt{n} \leq \|x\| \leq \frac{\sqrt{n}}{1-\varepsilon}\} \geq 1 - 2e^{-\frac{\varepsilon^2 n}{4}}$$

Since we fix $\varepsilon > \frac{4}{\sqrt{n}}$ we have

$$\mathbb{P}\{(1-\varepsilon)\sqrt{n} \leq \|x\| \leq \frac{\sqrt{n}}{1-\varepsilon}\} \geq 1 - 2e^{-4} \geq 0.97$$

We can use a conditional density scaled with $\frac{1}{0.97}$, but we prefer not to bother with this small detail, we'll directly use standard Gaussians to sample from the annulus

$$(1-\varepsilon)\sqrt{n} < \|x\| < (1+\varepsilon)\sqrt{n}$$

After the point x is sampled, we need to work on conditional expectations in Kac-Rice formula. We'll use [1, Thm 1.1] and standard importance sampling trick to simlify this expression. More precisely, let $A_i = B_i + \delta X_i$ where X_i are $n \times n$ Gaussian matrices with independent entries with mean zero and variance 1. These A_i can be sampled using a standard random matrix generator. However, these random do not satisfy the assumption in

$$\mathbb{E}\left(|det(Z^{'}(x))| \mid Z(x) = (c_1, c_2, \ldots, c_n)\right)$$

To remedy this situation, we pick a coordinate x_i of the sampled vector $(x_1, x_2, \ldots, x_n)$. To make sure x_i is neither too large or too small we can pick the coordinate with median magnitude among $|x_1|, |x_2|, \ldots, |x_n|$. After this we look at our equations:

$$\|A_1 x\|^2 = c_i \Leftrightarrow a_{i,i}^{(1)} x_i^2 + x_i(\sum_{j \neq i}^{n} x_j a_{i,j}^{(1)}) + \sum_{j,k \neq i} a_{j,k}^{(1)} x_j x_k = c_i$$

$$\|A_1 x\|^2 = c_i \Leftrightarrow -\sum_{j=1}^{n} a_{i,j}^{(1)} x_j x_i^{-1} - \sum_{j,k \neq i} a_{j,k}^{(1)} x_j x_k x_i^{-2} + c_i x_i^{-2} = a_{i,i}^{(1)} \tag{21}$$

For ease of notation, we set

$$g(A_j, c_j, x) := -\sum_{j=1}^{n} a_{i,j}^{(j)} x_j x_i^{-1} - \sum_{j,k \neq i} a_{j,k}^{(j)} x_j x_k x_i^{-2} + c_i x_i^{-2}$$

assuming the coordinate i is already decided. We will also use $g(B_j, c_j, x)$ to denote

$$g(B_j, c_j, x) = -\sum_{j=1}^{n} b_{i,j}^{(j)} x_j x_i^{-1} - \sum_{j,k \neq i} b_{j,k}^{(j)} x_j x_k x_i^{-2} + c_i x_i^{-2}$$

So, if we condition $a_{i,i}^{(1)}$ on all the rest of the entries of the random matrix A_1 using Equation (21) we can ensure $\|A_1 x\|^2 = c_1$ is satisfied. So we do the following update:

$$a_{i,i}^{j} \leftarrow g(A_j, c_j, x)$$

for all $j = 1, 2, \ldots, n$. In this case, the ratio of densities in importance will be given by the update from already sampled $a^(1)_{i,i}$ to $g(A_1, c_1, x)$, that is:

$$e^{-\frac{(g(A_1,c_1,x)-g(B_1,c_1,x))^2}{2}} e^{\frac{(a^1_{i,i}-b^1_{i,i})^2}{2}}$$

All in all, our density update for the entire system in Kac-rice formula is

$$p(G(x)) := \prod_{j=1}^{n} e^{-\frac{\left(g(A_j,c_j,x)-g(B_j,c_j,x)\right)^2}{2}} e^{\frac{(a^j_{i,i}-b^j_{i,i})^2}{2}}$$

Now let us define $G = (g(A_1, c_1, x), g(A_2, c_2, x), \ldots, g(A_n, c_n, x))$ and let $D_x G$ denote the Jacobian matrix of this system of equations evaluated at x. Theorem 1.1. from [1] shows that we then have the following equality:

$$\mathbb{E}\left(|det(Z^{'}(x))| \mid Z(x) = (c_1, c_2, \ldots, c_n)\right) = \int |\det(D_x G)|\, p(G(x))\, \mu_1 \mu_2 \ldots \mu_n \tag{22}$$

where $\mu_1, \mu_2, \ldots, \mu_n$ denotes the independent Gaussian distribution that is used to sample $A_1, A_2, \ldots, A_n$ initially.

Remark 5.1. *Strictly speaking* [1, Thm 1.1] *shows Equation* (22) *holds true for the case of uniform distribution on a bounded interval rather than Gaussian distribution that we've employed, however, the proof can be repeated verbatim to obtain the above inequality. We do not indulge into this technical exercise for the sake of not adding more technicality, but a skeptical reader is welcome to do the exercise.*

So, our Kac-Rice formula now looks like:

$$\int_{\mathbb{R}^n} \mathbb{E}\left(|det(Z^{'}(x))| \mid Z(x) = (c_1, c_2, \ldots, c_n)\right)\, dx = \int_{\mathbb{R}^n} \int |\det(D_x G)|\, p(G(x))\, \mu_1 \mu_2 \ldots \mu_n\, dx$$

5.0.3. *Vanilla Monte-Carlo.* So, our vanilla Monte-Carlo approximation to $\mathbb{E}N(\mathcal{B}, c, \delta)$ where $0 < \delta < \frac{1}{n}$ will go as follows:

(1) Sample $x_1, x_2, \ldots, x_N \in \mathbb{R}^n$ independently from standard Gaussian measure on $\mathbb{R}^n$. Here N will be determined by the user, it will determine the quality of approximation.
(2) For every sampled vector x_k, sample M tuples of random matrices

$$(A_t)_{t=1}^{M} = \left((A_1^t, A_2^t, \ldots, A_n^t)\right)_{t=1}^{M}$$

using standard Gaussian distribution, and compute the following incremental sum

$$\frac{1}{M} \sum_{t=1}^{M} |\det(D_{x_k} G(A_t, x_k, c_t))|\, p(G(A_t, c_t, x_k))$$

(3) Return the following as our estimate to Kac-Rice formula

$$\frac{1}{N} \sum_{k=1}^{N} \left(\frac{1}{M} \sum_{t=1}^{M} |\det(D_{x_k} G(A_t, x_k, c_t))|\, p(G(A_t, c_t, x_k)) \right)$$

5.0.4. *Hamiltonian Monte-Carlo.* After lessons learned from trying vanilla approach, we are likely going to spice it up with Hamiltonian Monte-Carlo here.

6. Direct Optimization in Value Space for Finding Extremal Equations

7. Actor-Critic Methods for Finding Extremal Equations

References

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